Experiment 1

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Title: Installation and Configuration of Anaconda Navigator, Python, TensorFlow, Keras and Gymnasium

Software Required:

- Anaconda Navigator
- Python
- TensorFlow
- Keras
- Gymnasium

AIM

To install and configure Anaconda Navigator, Python, TensorFlow, Keras, and Gymnasium for developing Reinforcement Learning applications.

THEORY

Anaconda is a Python distribution used for data science and machine learning. TensorFlow and Keras are libraries used to build deep learning models, while Gymnasium provides environments for developing and testing Reinforcement Learning algorithms.

PROCEDURE

1. Download and install **Anaconda Navigator**.
2. Launch Anaconda Navigator.
3. Create a new Python environment (optional).
4. Open **Anaconda Prompt**.
5. Install TensorFlow.
6. Install Keras , Install Gymnasium.
7. Verify that all libraries are installed successfully.
8. Run a simple Python program to import the libraries.
9. Observe the output.

CODE:

```
# Install Gymnasium
!pip -q install gymnasium

# Import Libraries
import tensorflow as tf
import keras
import gymnasium as gym
import matplotlib.pyplot as plt

# Display Versions
print("TensorFlow Version :", tf.__version__)
print("Keras Version      :", keras.__version__)
print("Gymnasium Version  :", gym.__version__)

# Create Environment
env = gym.make("CartPole-v1")
observation, info = env.reset()

print("\nEnvironment Created Successfully!")
print("Initial Observation:", observation)

# Extract Version Numbers
tf_version = float(".".join(tf.__version__.split(".")[:2]))
keras_version = float(".".join(keras.__version__.split(".")[:2]))
gym_version = float(".".join(gym.__version__.split(".")[:2]))

# Comparison Graph
libraries = ["TensorFlow", "Keras", "Gymnasium"]
versions = [tf_version, keras_version, gym_version]

plt.figure(figsize=(8,5))
bars = plt.bar(libraries, versions)

plt.title("Comparison of Installed Library Versions")
plt.xlabel("Libraries")
plt.ylabel("Version")

# Display version values on bars
for bar, version in zip(bars, versions):
    plt.text(bar.get_x() + bar.get_width()/2,
```

```
            bar.get_height() + 0.05,
            str(version),
            ha='center')

plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()

env.close()
```

OUTPUT:

```
TensorFlow Version : 2.20.0
Keras Version      : 3.13.2
Gymnasium Version  : 1.3.0

Environment Created Successfully!
Initial Observation: [-0.01586977  0.01386712 -0.00241283 -
0.00816068]
```